

Data Quality & Audit Report

1. Overview

This report documents the data quality assessment performed on the raw D2C customer dataset (snapshot date: 2025-09-30). Several intentional anomalies and standard data hygiene issues were identified that must be addressed prior to feature engineering and modeling.

2. Identified Data Quality Issues

A. Data Leakage Risks (Critical)

- **Issue:** The orders.csv file contains transactions with order_date values strictly greater than the snapshot date (2025-09-30).
- **Impact:** Using these post-snapshot orders to calculate RFM (Recency, Frequency, Monetary) metrics or other features will result in target leakage, artificially inflating model performance while rendering it useless in production.
- **Treatment:** All rows where order_date > '2025-09-30' must be filtered out entirely before any aggregations are performed.

B. Duplicate / Duplicate-Like Records

- **Issue:** In orders.csv, several order_id values end with a _DUP suffix, simulating real-world upstream system deduplication issues.
- **Impact:** Double-counting these orders will inflate a customer's total lifetime value (LTV) and order frequency metrics.
- **Treatment:** Strip the _DUP suffix and drop duplicate records based on order_id, or sort by order_date and keep the first occurrence to ensure clean transaction logs.

C. Missing Values

- **Issue 1:** customers.csv has missing values in loyalty_tier and skin_type.
 - *Treatment:* Impute missing loyalty_tier with a "None" or "Unenrolled" category. Treat missing skin_type as "Unknown" since a customer might simply have skipped that profile question.
- **Issue 2:** orders.csv is missing rating for approximately 80 rows.
 - *Treatment:* Fill missing ratings with the customer's historical average rating, the global median rating, or isolate them into an "Unrated" category before computing customer-level averages.

D. Outliers & Anomalies

- **Issue:** orders.csv contains extreme outlier values in the gross_amount column (e.g., values up to ₹24,789).
- **Impact:** Outliers can skew average order value (AOV) calculations and affect

distance-based machine learning algorithms.

- **Treatment:** Apply a capping strategy (e.g., 99th percentile winsorization) or investigate if these represent wholesale/B2B purchases that should be segmented differently.

3. Join / Key Integrity

- All files successfully join on customer_id using a Left Join originating from the primary customers.csv table. There are no orphan records in the event tables that do not map back to a valid customer profile.